



THE CANADIAN CHEMISTRY CONTEST 2022
PART A – MULTIPLE CHOICE QUESTIONS (60 minutes)

All contestants should attempt this part of the contest before proceeding to Part B and/or Part C.

The only reference material allowed is the CIC/CCO Periodic Table provided. You must complete answers on the Scantron Sheet provided.

Students may use a scientific calculator. No phones or communication devices are allowed.

- 1) Which of the following substances does not require the following WHMIS symbol (*oxidizer*)?



A) Cl_2 B) O_2 C) NaClO D) $\text{Pb}(\text{NO}_3)_2$ **E) H_2**

- 2) Which combination of atoms will form stable compounds with the following molecular geometries: bent, see-saw, octahedral?

A) sulfur and fluorine D) two of the combinations of atoms
B) xenon and fluorine E) all three combinations of atoms
C) bromine and fluorine

- 3) Consider the two species He and Li^+ . Which statement best describes the comparison of their ionization energies and radii?

A) Li^+ has a smaller radius and greater ionization energy than He
B) Li^+ has a larger radius and lower ionization energy than He
C) He has a smaller radius and greater ionization energy than Li^+
D) He has a smaller radius and lower ionization energy than Li^+
E) Li^+ has a smaller radius and lower ionization energy than He

- 4) A chemist combines 10.0 mL of 0.25 mol L^{-1} antimony (III) nitrate, $\text{Sb}(\text{NO}_3)_3$, with 10.0 mL of 0.35 mol L^{-1} sodium sulfide, Na_2S . What is the theoretical yield of the precipitate?

A) 0.40 g B) 0.42 g C) 0.60 g D) 0.85 g E) 3.6 g

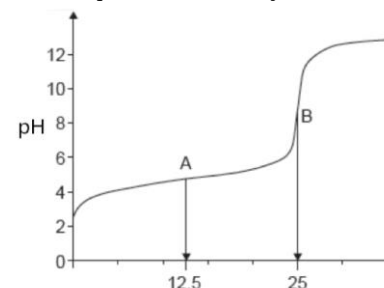
- 5) 1.0 L of 0.010 M NaOH is added to 0.50 L of water. What is the pH of the solution?

A) 2.00 B) 2.18 C) 9.18 **D) 11.82** E) 12.18

- 6) Chemistry students learn that, under many environmental conditions, the behaviour of gases can be approximated using the Ideal Gas Law. However, in practice, many gases deviate somewhat from ideal gas behaviour. Using your knowledge of intermolecular forces, determine which of the following gases should have the lowest pressure when all other variables remain constant.

A) Cl_2 B) Xe C) CH_4 **D) NH_3** E) CO

- 7) The graph represents the titration of 25.00 mL of 0.100 mol L^{-1} aqueous HA with 0.100 mol L^{-1} aqueous sodium hydroxide.



Volume (mL) of 0.100 mol L^{-1} NaOH

Which of the following statements is false?

- A) HA is not 100% ionized in water.
B) At point A, other than H_2O and Na^+ , the major species are HA and A^- .
C) Adding silver nitrate at point B will not affect the pH of the solution.
D) At point A, the pK_a of HA is equal to the pH of the solution.
E) The acid dissociation constant of HA is less than one.

- 8) If 2.38 g of a gas are released into an evacuated 2.00 L vessel at 22.5°C and the pressure inside the container rises to 104.4 kPa, what is a possible identity of the gas? The ideal gas constant is 8.314 kPa L mol⁻¹ K⁻¹; assume ideal behaviour of the gas.

A) CO₂ **B) N₂** C) F₂ D) Xe E) CH₄

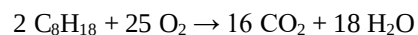
- 9) If you prepared 0.25 mol L⁻¹ aqueous solutions of each of three electrolytes, which list of three would all have a pH > 7?

A) NH₄NO₃, NaOH, Na₂CO₃ **D) Na₃PO₄, NaF, NaClO**
 B) Al(NO₃)₃, NaCH₃COO, NaHCO₃ E) Na₂S, Ca(NO₃)₂, Fe(NO₃)₃
 C) NaNO₃, NaClO₄, Na₂SO₄

- 10) Biodiesels are an alternative to fossil fuels. Unlike fossil fuels, biodiesels are derived from recently grown plants that consumed greenhouse gases while they were growing. A 0.435 g sample of C₁₄H₂₈O₂ biodiesel fuel combusts in a bomb calorimeter containing 500.0 g of water initially at 20.00°C. After the combustion, the final temperature of the water is 49.7°C. The calorimeter has a heat capacity of 390 J °C⁻¹ and the specific heat capacity of water is 4.184 J g⁻¹ °C⁻¹. What is the heat of combustion of the biodiesel?

A) - 7.37 x 10¹ kJ mol⁻¹ **B) - 3.87 x 10⁴ kJ mol⁻¹** C) - 6.21 x 10¹ kJ mol⁻¹
 D) - 1.15 x 10³ kJ mol⁻¹ E) - 5.05 x 10³ kJ mol⁻¹

- 11) Many cars expose fuel to 1.50 L of atmospheric air every 2 revolutions. Assume atmospheric air contains 21.0 % oxygen gas by volume. If the combustion reaction is standardized to SATP conditions and uses an excess amount of pure octane (C₈H₁₈), calculate the volume of carbon dioxide produced in 30 seconds when the engine idles at 1500 revolutions per minute. Use the following balanced chemical reaction:



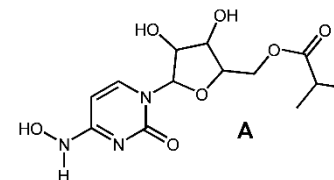
A) 360 L B) 263 L C) 184 L D) 118 L **E) 75.6 L**

- 12) How many chloride ions are present in 35 mL of a 0.70 mol L⁻¹ solution of CaCl₂ (aq)?

A) 0.0245 B) 49 C) 1.5 x 10²² **D) 3.0 x 10²²** E) 6.0 x 10²³

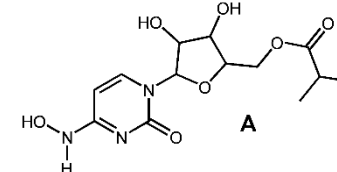
- 13) The ongoing COVID-19 pandemic has led to the development of an antiviral drug compound (**A**) which was approved for medical use in late 2021. The structure of compound **A** is shown below. How many carbon atoms and hydrogen atoms are there in this compound?

A) 12 carbon and 18 hydrogen atoms
 B) 12 carbon and 19 hydrogen atoms
 C) 13 carbon and 18 hydrogen atoms
D) 13 carbon and 19 hydrogen atoms
 E) 13 carbon and 20 hydrogen atoms



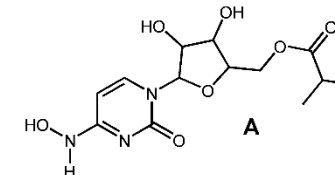
- 14) Which list of functional groups are **all** present in compound **A**?

A) alcohol, carboxylic acid, amide
 B) amine, ether, carboxylic acid
 C) alcohol, ketone, amine
 D) ether, ketone, amide
E) ether, alcohol, ester



- 15) Two compounds structurally related to **A** were synthesized and subjected to medical testing. These two new compounds were labelled **B** and **C**. In the structure of compound **B**, the nitrogen atoms in compound **A** are replaced by phosphorus atoms. In the of compound **C**, the oxygen atoms in compound **A** are replaced by sulfur atoms. What is the difference in the molecular weight of compounds **B** and **C**?

A) 29.4
 B) 45.5
C) 61.5
 D) 78.6
 E) 94.8



- 16) Which of the following elements forms a stable +4 ion with the following shorthand electron configuration: [noble gas] (n-1) d³?

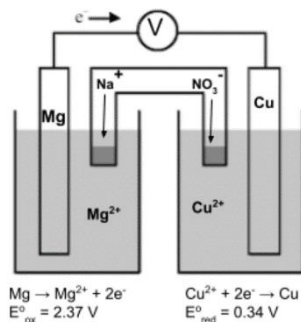
A) Mn B) Co C) Pb D) V E) Ti

- 17) For the hypothetical reaction, $2A + 3B \rightarrow A_2B_3$, when the initial $[A] = 0.100 \text{ mol L}^{-1}$ and the initial $[B] = 0.100 \text{ mol L}^{-1}$, the reaction rate is $1.4 \times 10^{-1} \text{ mol L}^{-1} \text{ s}^{-1}$. Which experiment in the table below would be consistent with the rate law, $\text{rate} = k[A][B]^2$?

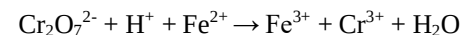
Experiment	Initial [A] (mol L ⁻¹)	Initial [B] (mol L ⁻¹)	Reaction Rate (mol L ⁻¹ s ⁻¹)
A)	0.200	0.200	4.48×10^0
B)	0.200	0.300	1.68×10^0
C)	0.300	0.200	1.68×10^0
D)	0.300	0.300	1.13×10^1
E)	0.200	0.400	2.24×10^0

- 18) Which statement about this electrochemical cell setup is **incorrect**.

A) The voltmeter reading would be 2.71 V
 B) The direction of electron movement is correct in the diagram
 C) The direction of the salt bridge ion movement is correct in the diagram
 D) The cathode half-cell reaction is correct in the diagram
 E) The magnesium electrode would become lighter as the cell operated

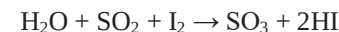


- 19) When the following redox reaction is balanced with the lowest possible integer coefficients, what is the sum of the coefficients of all reactants and products?



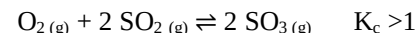
A) 6 B) 18 C) 26 D) 30 E) 36

- 20) A biochemist performs a Karl Fischer titration to calculate the water content of a 26.0 mg unknown sample of a powder. Assume all of the water in the unknown sample reacts. If 11.2 mg of I₂ were consumed during the titration, use the following balanced equation for the Karl Fischer reaction to determine the percent water content by mass in the unknown sample.



A) 1.70 % B) 3.06 % C) 4.42 % D) 6.12 % E) 21.5 %

- 21) 1.0 mol of sulfur dioxide and 1.0 mol of oxygen gas react and establish an equilibrium as follows:



Which addition of reactants to the equilibrium reaction system will cause the greatest increase in the amount of sulfur trioxide? Assume both the volume and the temperature of the reaction mixture remain constant.

	Moles of O _{2(g)} added	Moles of SO _{2(g)} added
A)	0.8	1.2
B)	0.9	1.1
C)	1.0	1.0
D)	1.1	0.9
E)	1.2	0.8

- 22) A 25.00 mL sample of 0.800 M sodium acetate was titrated to equivalence with 15.00 mL of hydrochloric acid solution. Which of the following calculations represents the pH of the final solution?

(acetic acid $K_a = 1.8 \times 10^{-5}$)

- A) $1.8 \times 10^{-5} = \frac{x^2}{(0.800 - x)}$
- B) $-\log((0.800)(\frac{0.02500}{(0.02500+0.01500)}))$
- C) $-\log\sqrt{(1.8 \times 10^{-5})(0.800)(\frac{0.02500}{(0.02500+0.01500)})}$
- D) $14 - (-\log\sqrt{(\frac{1.8 \times 10^{-5}}{1 \times 10^{-14}})(0.800)(\frac{0.02500}{(0.02500+0.01500)})})$
- E) $-\log\sqrt{(1.8 \times 10^{-5})(0.800)}$

- 23) The green chemistry metric called *process mass intensity* (PMI) allows a chemist to calculate the total material used to synthesize a certain amount of desired product in a chemical reaction:

$$\text{PMI} = [(\text{mass of all input materials}) / (\text{actual mass of desired product})]$$

Consider the following laboratory procedure used to synthesize benzocaine, a local anesthetic used to treat oral ulcers:

0.0144 moles of 4-aminobenzoic acid hydrochloride (formula: $\text{C}_7\text{H}_8\text{ClNO}_2$) was dissolved in 20.0 mL of ethanol (density: 0.790 g/mL) and 1.1 mL concentrated H_2SO_4 (density: 1.82 g/mL) was added. The mixture was then rapidly stirred and heated vigorously for two hours. After cooling to room temperature, the reaction mixture was neutralized with 20.0 mL of a 10% aqueous sodium carbonate solution (density: 1.10 g/mL) and transferred to a separatory funnel. Extraction of the product was performed with 30.0 mL of dichloromethane (density: 1.325 g/mL) and the dichloromethane was subsequently dried with 6.0 g of anhydrous sodium sulfate. Removal of this drying agent and evaporation of the dichloromethane left 0.0102 moles of pure benzocaine (formula: $\text{C}_9\text{H}_{11}\text{NO}_2$).

The PMI for this reaction is

- A) 46.6 B) 48.8 C) 50.9 D) 51.2 E) 52.3

- 24) Consider the table below with the relative strength of acids and bases. Assuming all reactant species have the same concentration, which of the following equilibria would you expect to have the lowest K_{eq} value?

- A) $\text{CH}_3\text{COOH}(aq) + \text{NH}_3(aq) \rightleftharpoons \text{CH}_3\text{COO}^-(aq) + \text{NH}_4^+(aq)$
- B) $\text{H}_2\text{S}(aq) + \text{H}_2\text{O}(l) \rightleftharpoons \text{HS}^-(aq) + \text{H}_3\text{O}^+(aq)$
- C) $\text{HCl}(aq) + \text{H}_2\text{O}(l) \rightarrow \text{H}_3\text{O}^+(aq) + \text{Cl}^-(aq)$
- D) $\text{HF}(aq) + \text{H}_2\text{O}(l) \rightleftharpoons \text{H}_3\text{O}^+(aq) + \text{F}^-(aq)$
- E) $\text{HCN}(aq) + \text{NH}_3(l) \rightleftharpoons \text{CN}^-(aq) + \text{NH}_4^+(aq)$

Strongest Acid	HCl	Cl^-	Weakest Base
	H_2SO_4	HSO_4^-	
	H_3O^+	H_2O	
	H_2SO_3	HSO_3^-	
	HF	F^-	
	CH_3COOH	CH_3COO^-	
	H_2CO_3	HCO_3^-	
	H_2S	HS^-	
	HCN	CN^-	
	NH_4^+	NH_3	
Weakest Acid	H_2O	OH^-	Strongest Base

- 25) For a reaction that is spontaneous at -100°C but non-spontaneous at 200°C , what must be true about the reaction?

- A) The reaction is endothermic and $K_{eq} < 0$
- B) The reaction is exothermic and $K_{eq} > 0$
- C) The reaction is endothermic and decreases entropy
- D) The reaction is exothermic and increases entropy
- E) $\Delta H < 0$ and $\Delta S < 0$

End of Part A of the contest
Go back and check your work